ADVANCED COMPUTER NETWORKS-PCS223

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ME - CSE1

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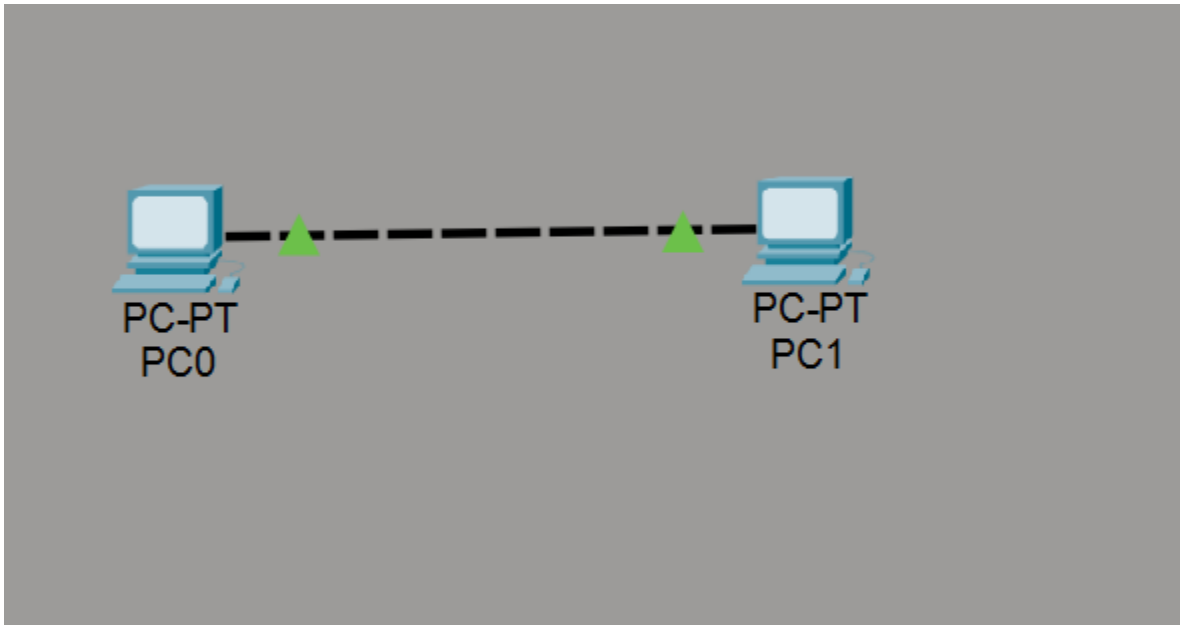
TIET, Patiala
2026

ASSIGNMENT 3

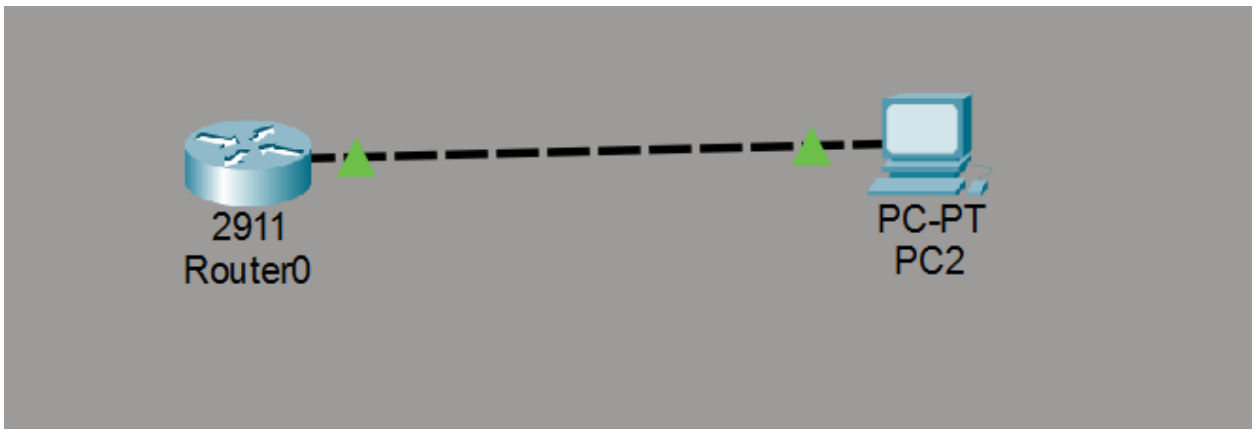
Ques1:

Sol-



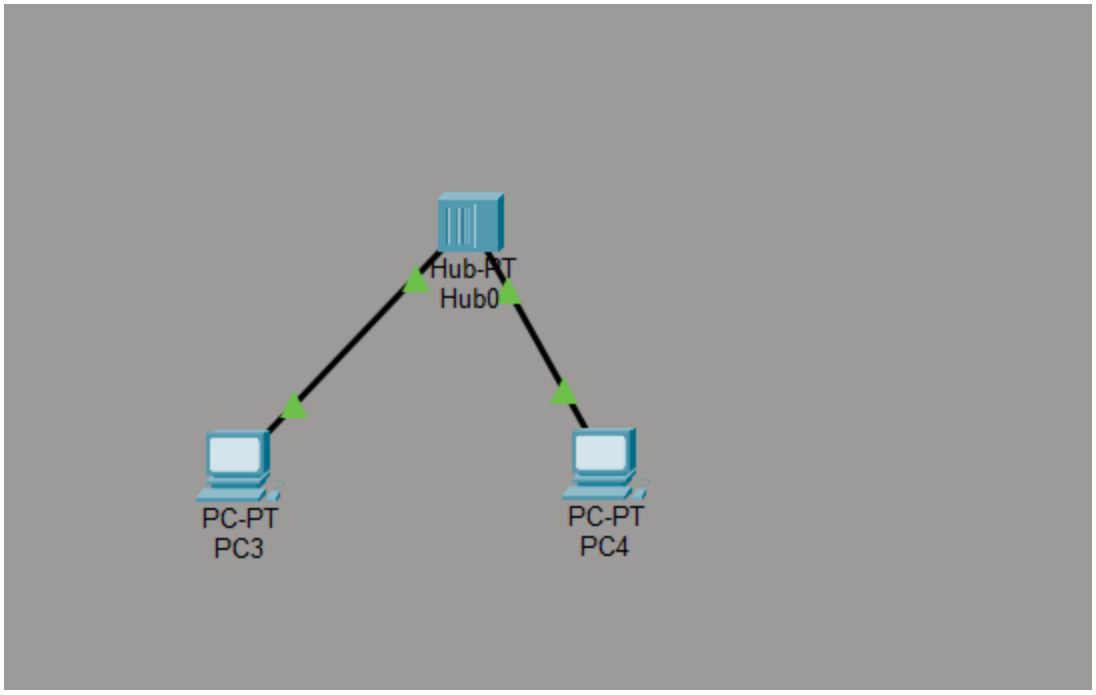
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)

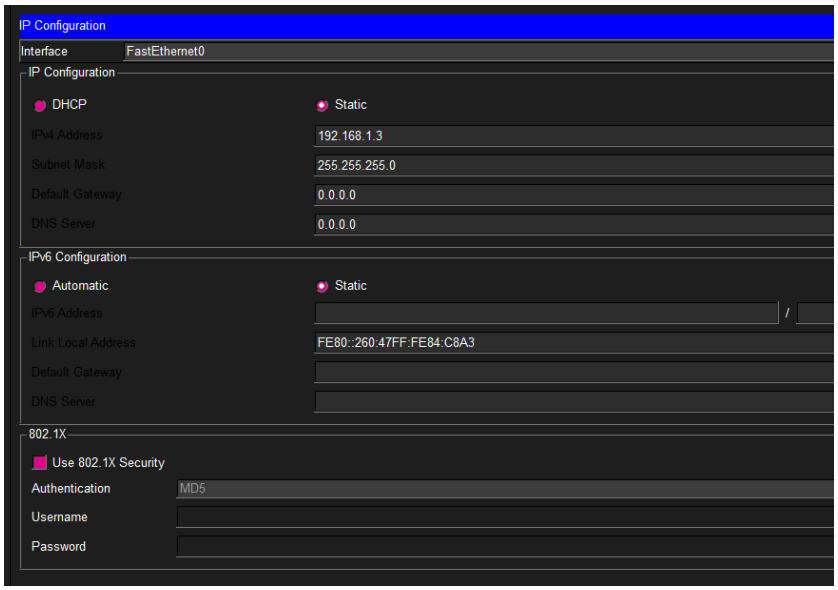
Ques 2:

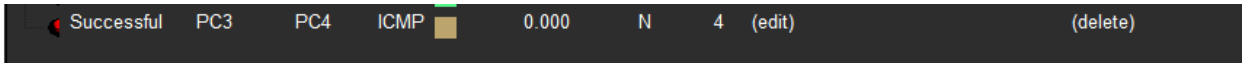


Successful	PC2	Router0	ICMP		0.000	N	1	(edit)	(delete)

Ques 3:


Hub-PT
Hub0
PC-PT
PC3
PC-PT
PC4

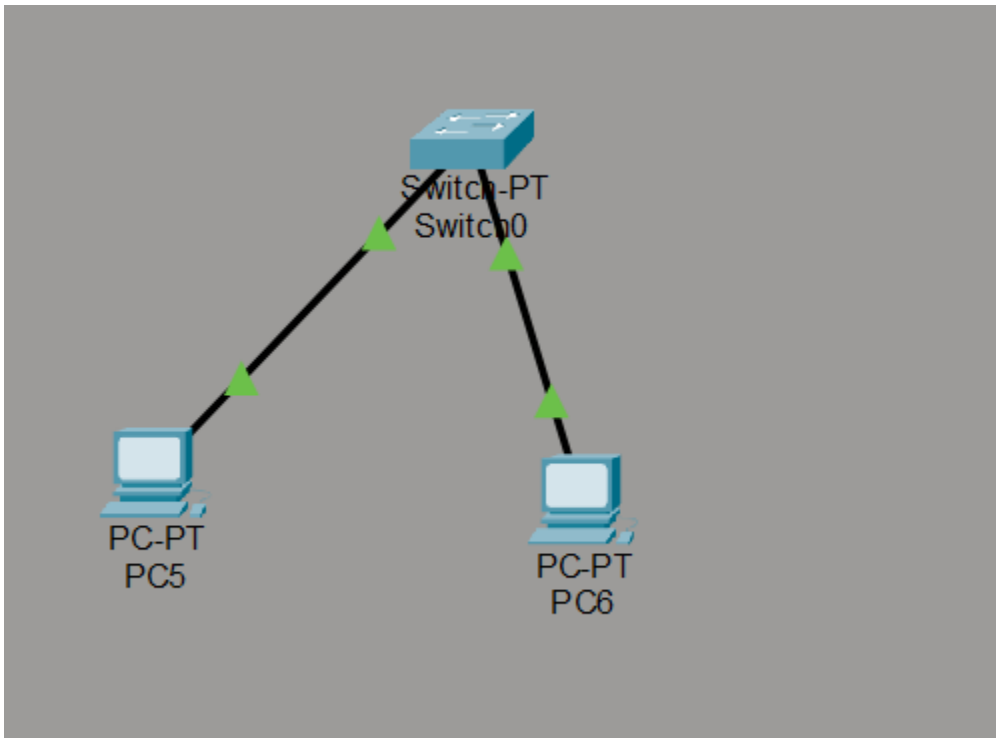

IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.3
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::260:47FF:FE84:C8A3
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

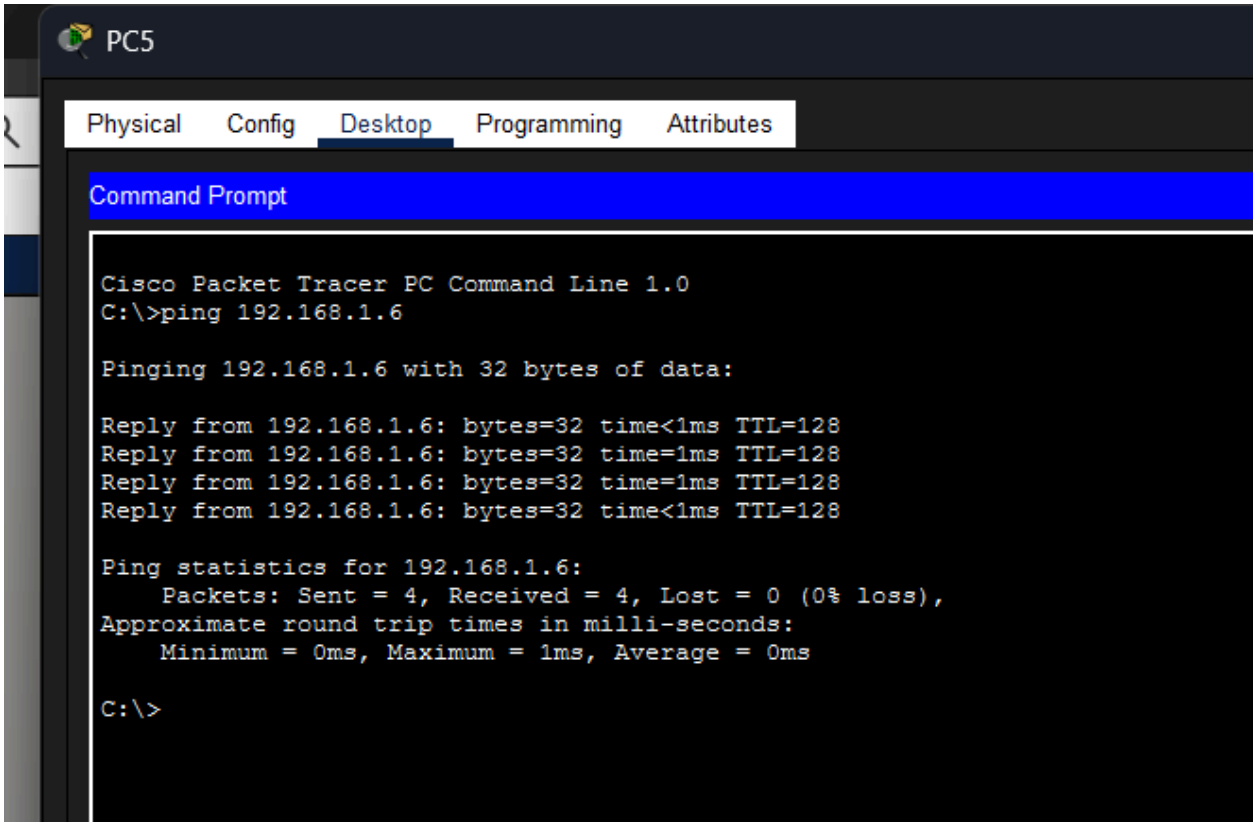

Successful PC3 PC4 ICMP 0.000 N 4 (edit) (delete)

```
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

\

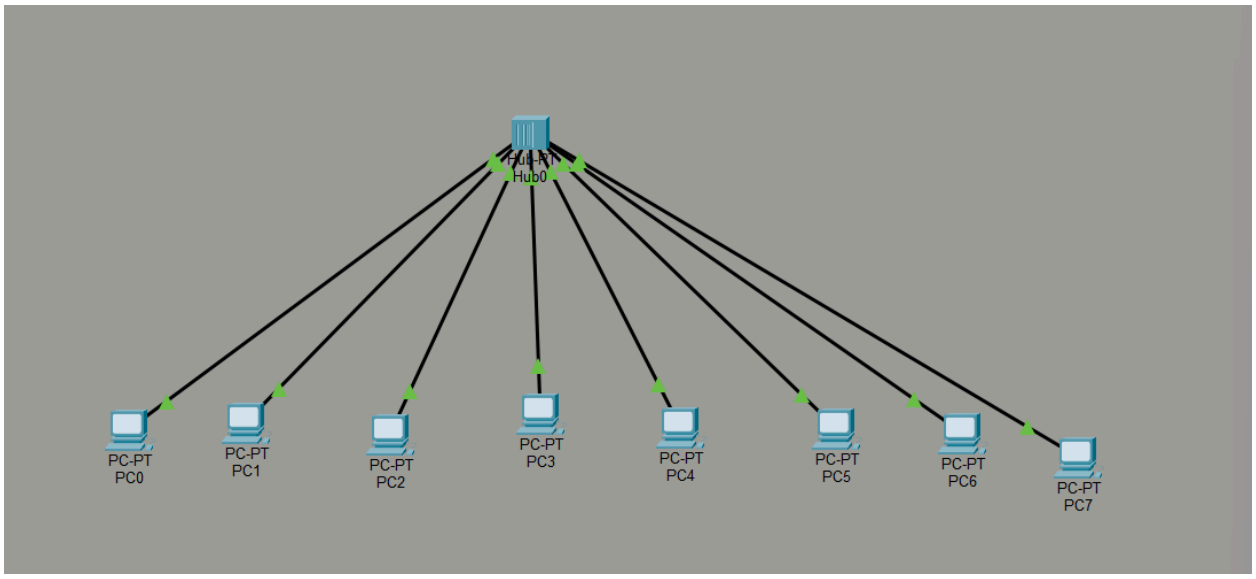
Ques 4:




PC5
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.6
Pinging 192.168.1.6 with 32 bytes of data:
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time=1ms TTL=128
Reply from 192.168.1.6: bytes=32 time=1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>

Ques 5:

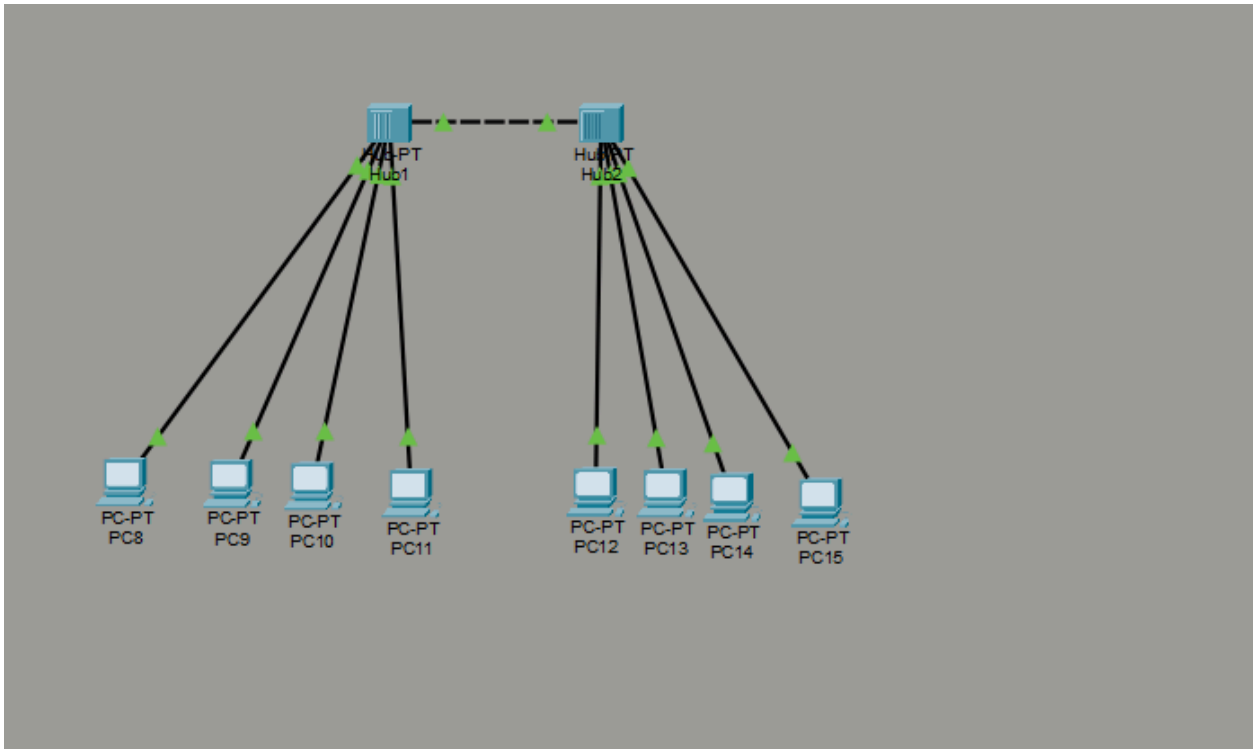
5(a)


Hub-PT
Hub0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5
PC-PT
PC6
PC-PT
PC7

Realtime Simulation

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC2	ICMP		0.000	N	4	(edit)	(delete)
	Successful	PC2	PC3	ICMP		0.000	N	5	(edit)	(delete)
	Successful	PC3	PC4	ICMP		0.000	N	6	(edit)	(delete)
	Successful	PC4	PC5	ICMP		0.000	N	7	(edit)	(delete)

5(b)



Realtime

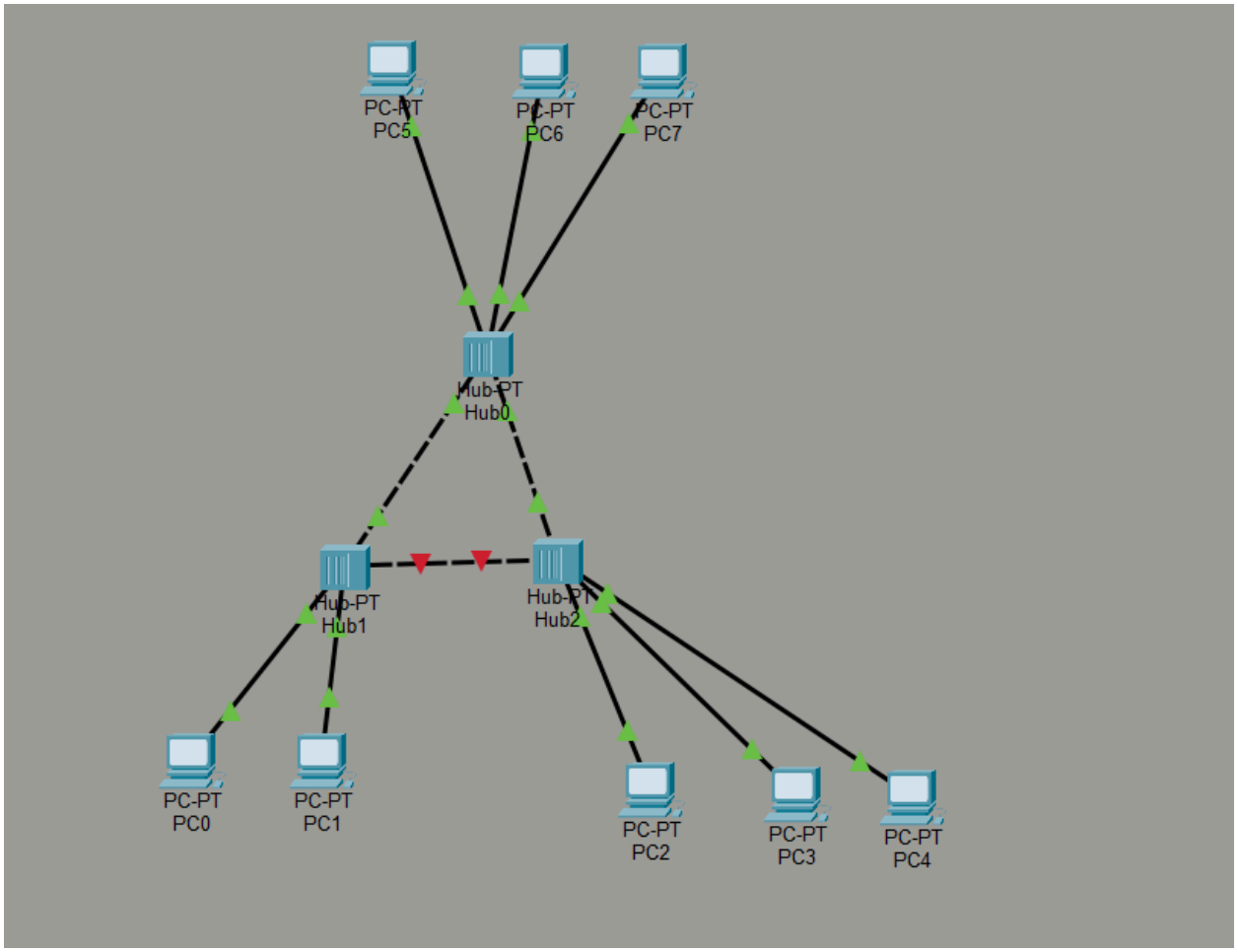
ario 0

Delete

ist Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC3	PC4	ICMP		0.000	N	6	(edit)	(delete)
	Successful	PC4	PC5	ICMP		0.000	N	7	(edit)	(delete)
	Successful	PC5	PC6	ICMP		0.000	N	8	(edit)	(delete)
	Successful	PC6	PC7	ICMP		0.000	N	9	(edit)	(delete)

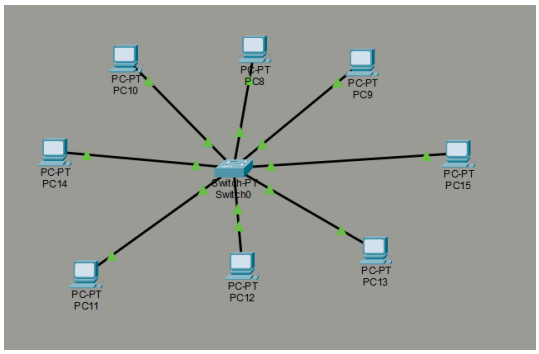
6.



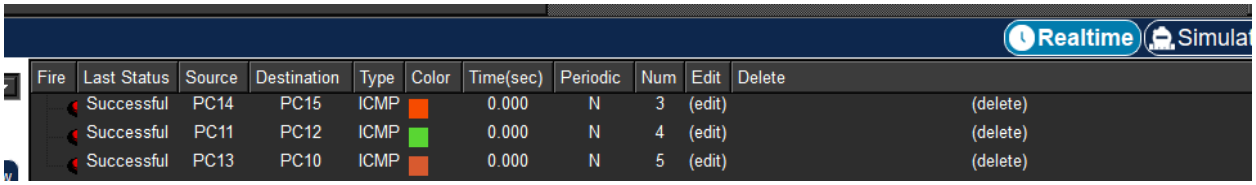
Realtime Simulati

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC7	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC0	PC1	ICMP		0.000	N	2	(edit)	(delete)

7.

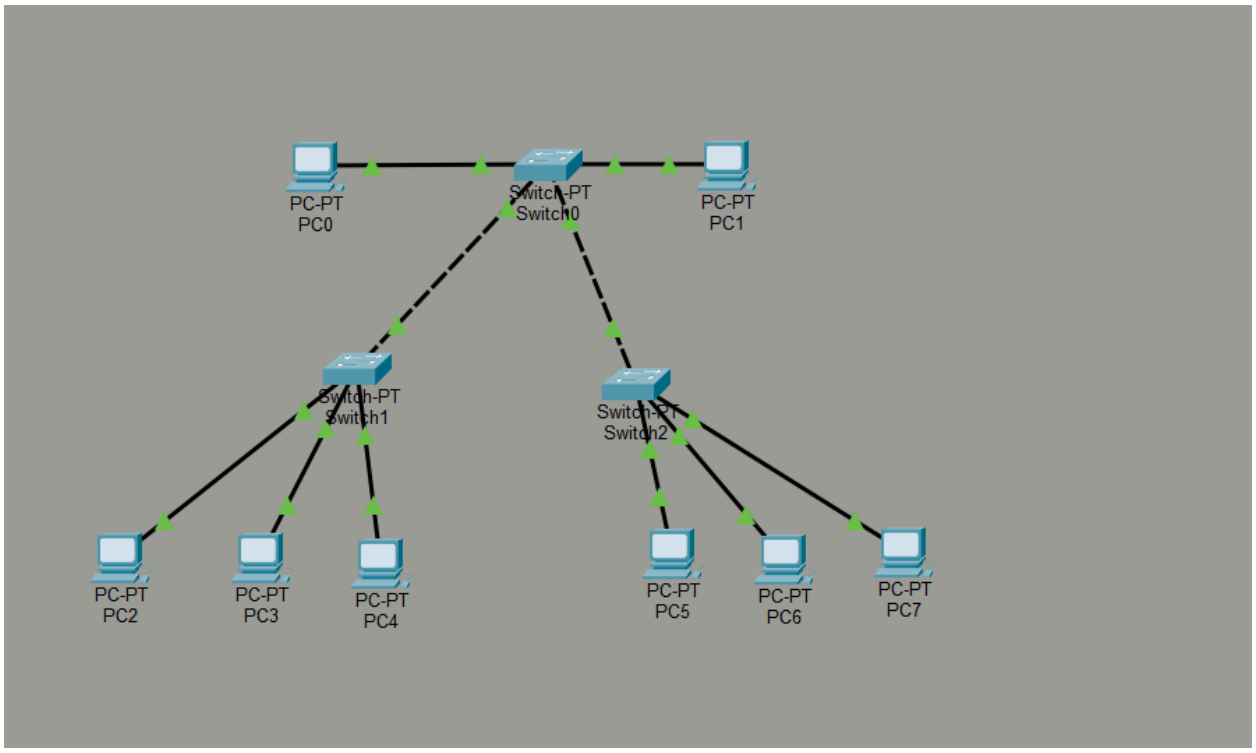


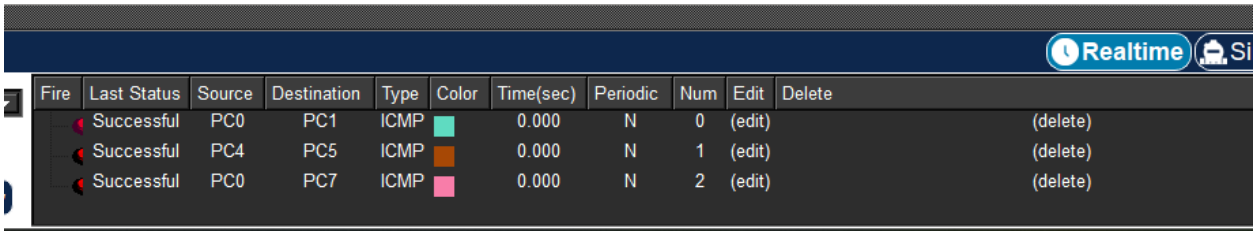
`



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC14	PC15	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC11	PC12	ICMP		0.000	N	4	(edit)	(delete)
	Successful	PC13	PC10	ICMP		0.000	N	5	(edit)	(delete)

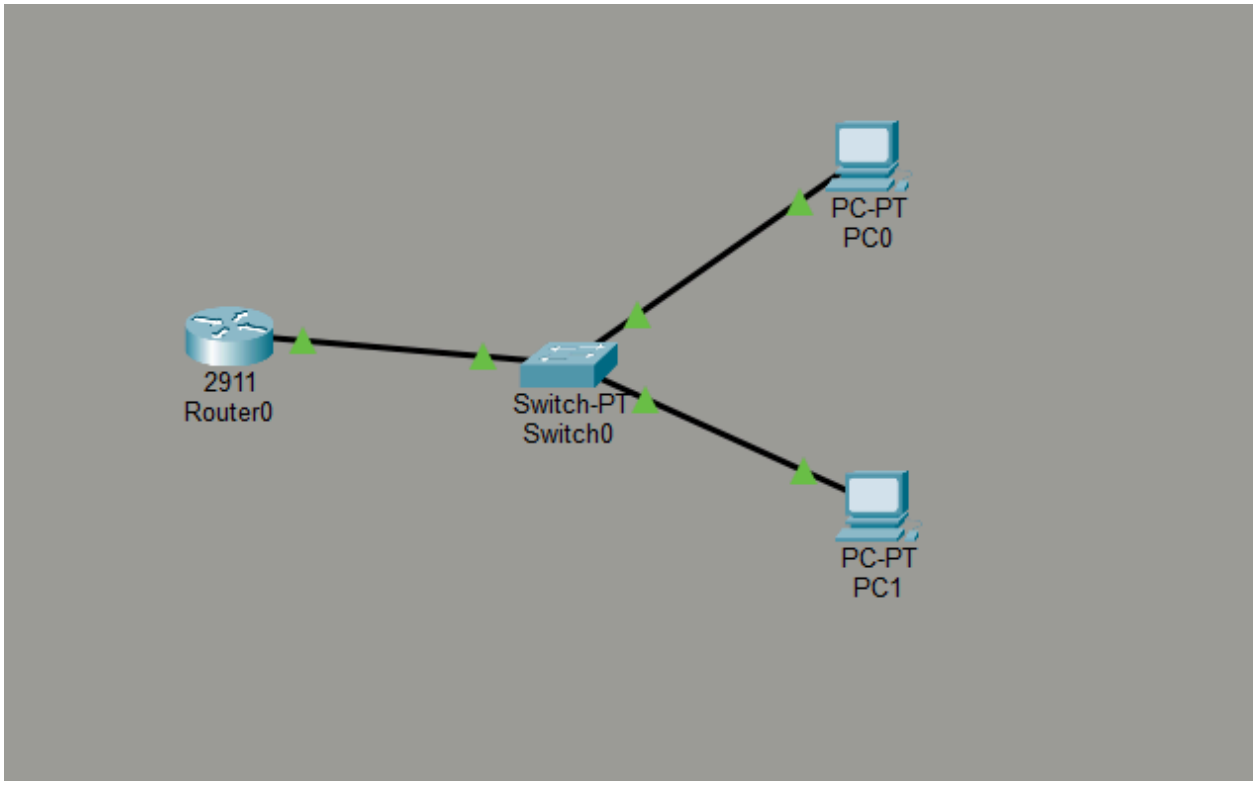
8.





Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC4	PC5	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC0	PC7	ICMP		0.000	N	2	(edit)	(delete)

ASSIGNMENT 4

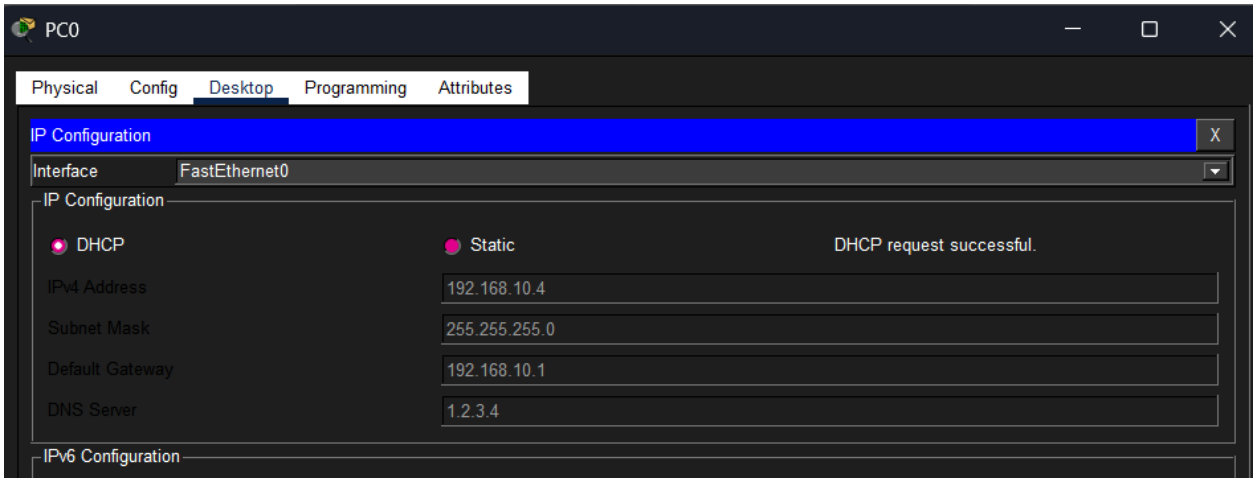


Router-

```
Router>enable
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip dhcp pool xyz
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 1.2.3.4
Router(dhcp-config)#exit
```

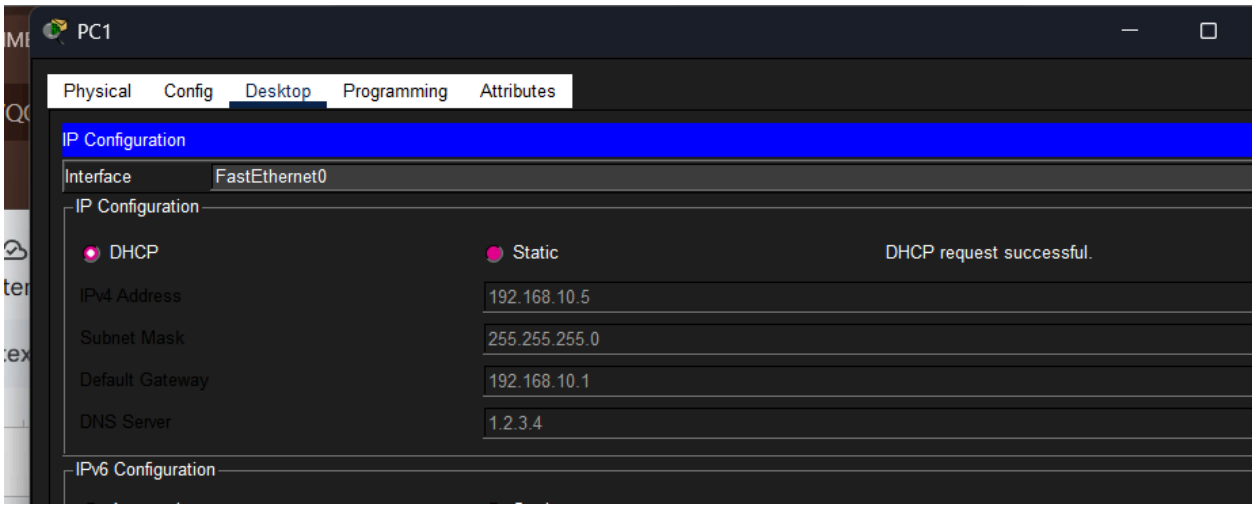
Pc0

-dhcp

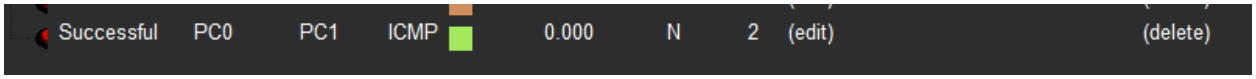


Pc1

-dhcp

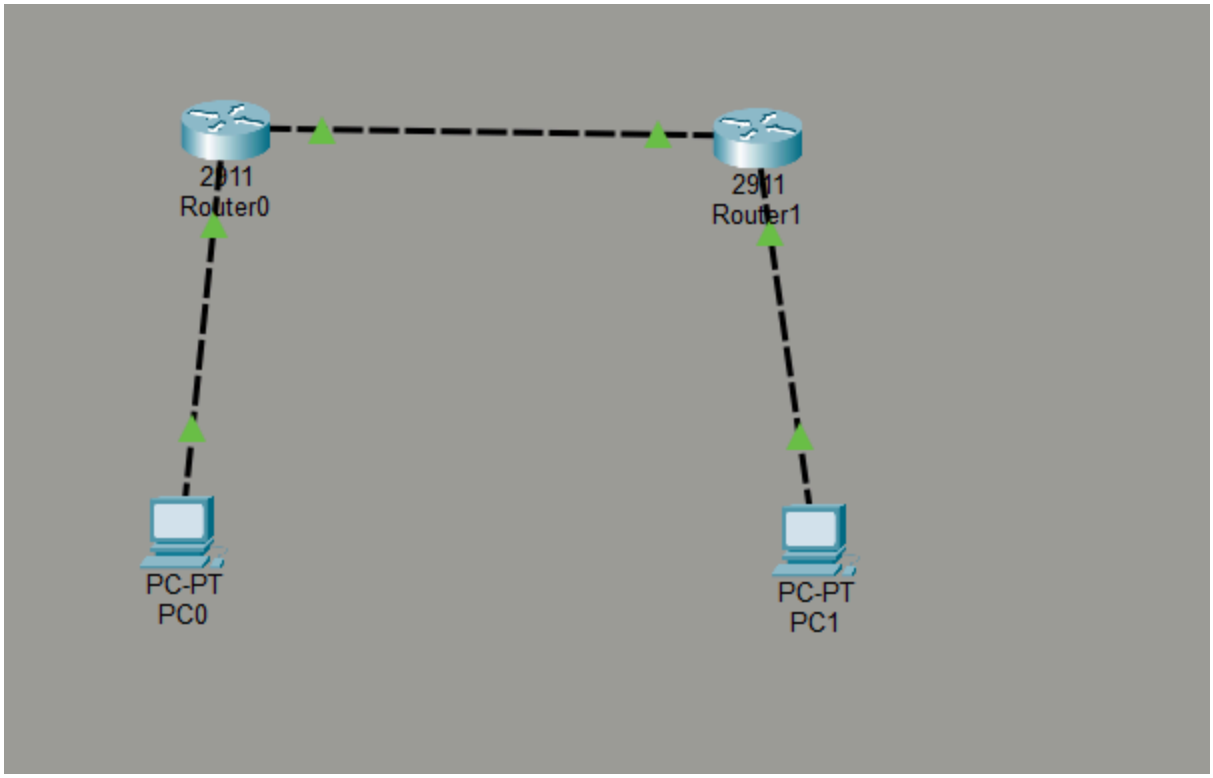


Output-



Assignment 5

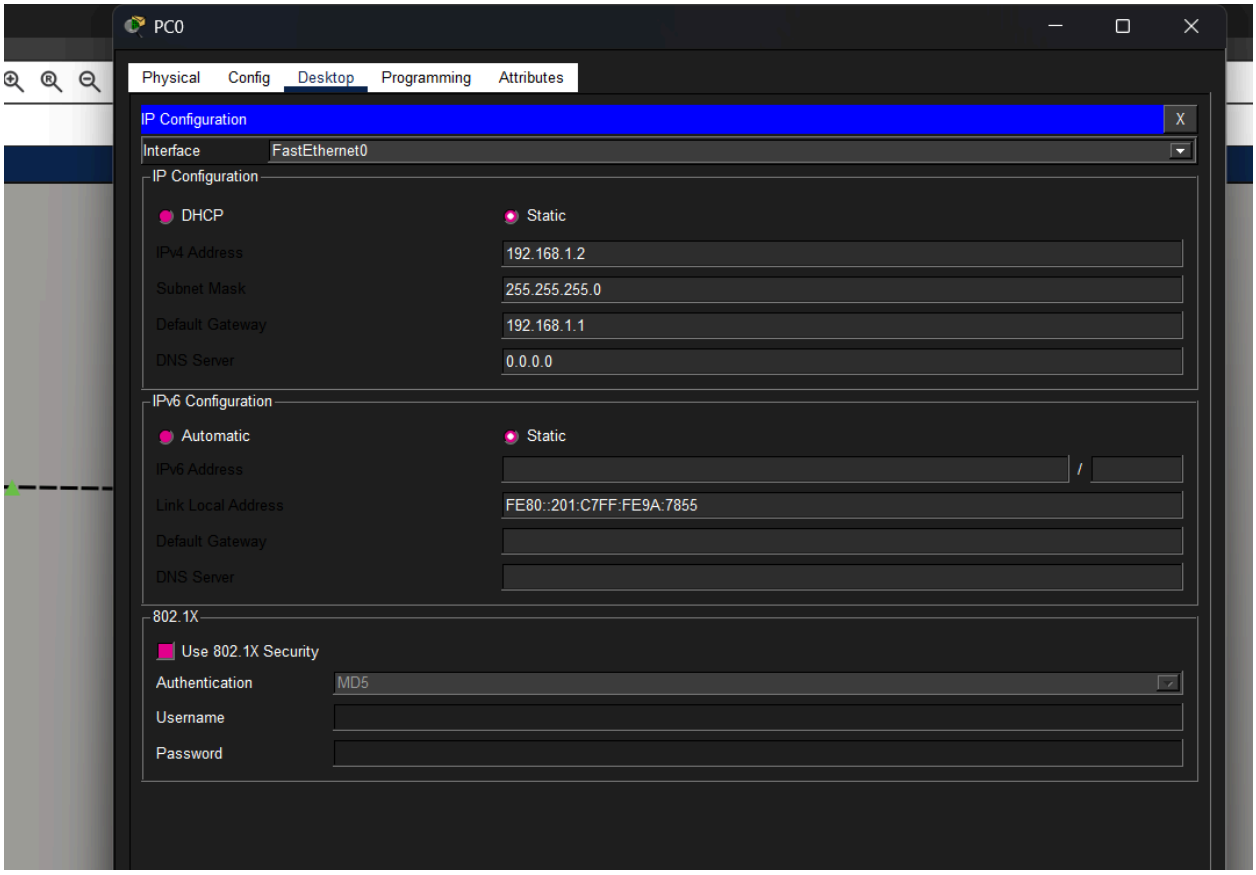
1-apr-2026- Example of 6th Assignment



```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet0/0
Router(config-if)#ip address 10.0.0.8
% Incomplete command.
Router(config-if)#ip address 10.0.0.8 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitethernet0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

```
Router>
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no autosummary
                             ^
% Invalid input detected at '^' marker.

Router(config-router)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#no auto summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Pc -0

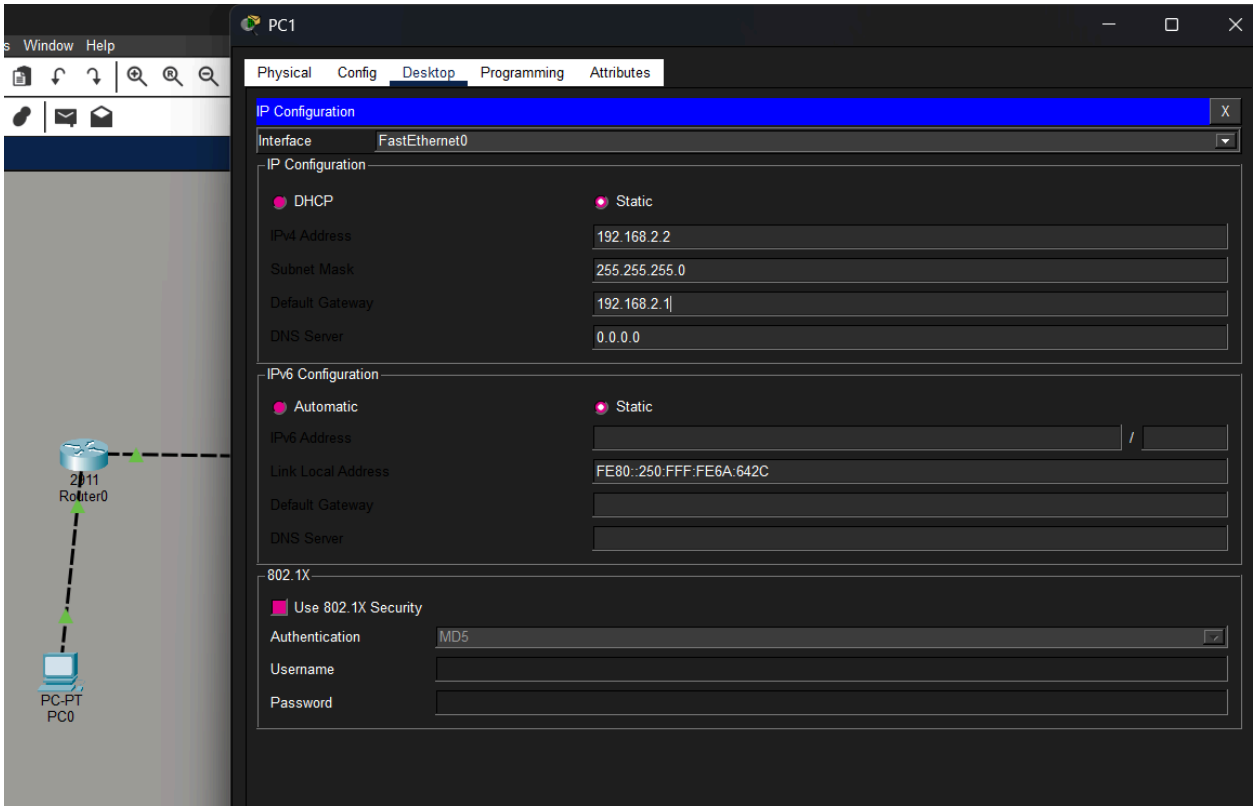

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:C7FF:FE9A:7855
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

Router 2

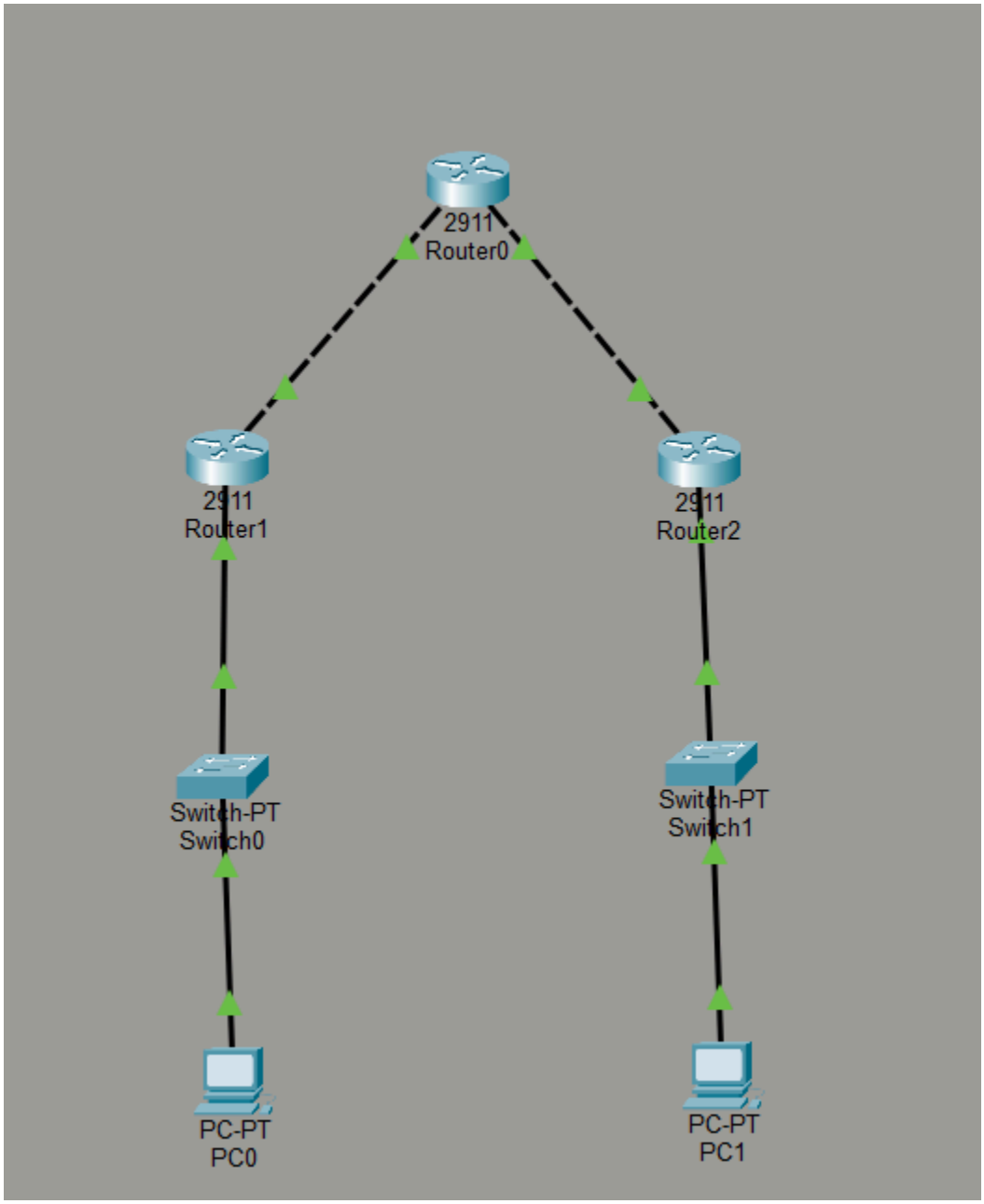
```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet0/0
Router(config-if)#ip address 10.0.0.9 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitethernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
```

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto summary
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto summary
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Pc 1


Window Help
PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::250:FFF:FE6A:642C
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
2911
Router0
PC-PT
PC0

Assignment 6



Router 0-

```
Press RETURN to get started!



Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 100.1.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface gig0/1
Router(config-if)#ip address 20.1.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 100.1.1.0
Router(config-router)#network 20.1.1.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 1-

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 100.1.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface gig0/1
Router(config-if)#ip address 172.16.2.1 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 100.1.1.0
Router(config-router)#network 172.16.0.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

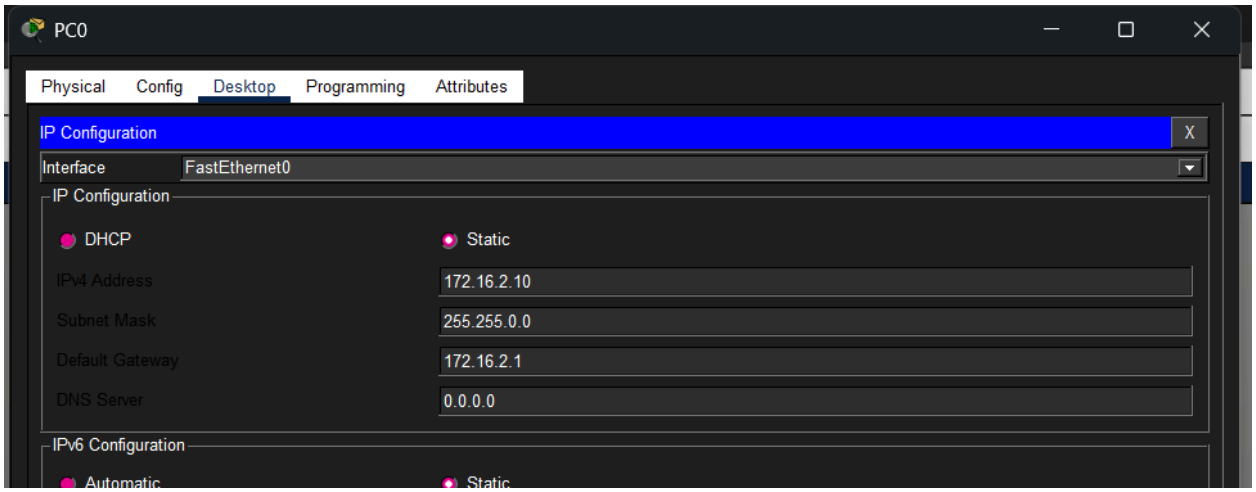
Copy

Router 2-

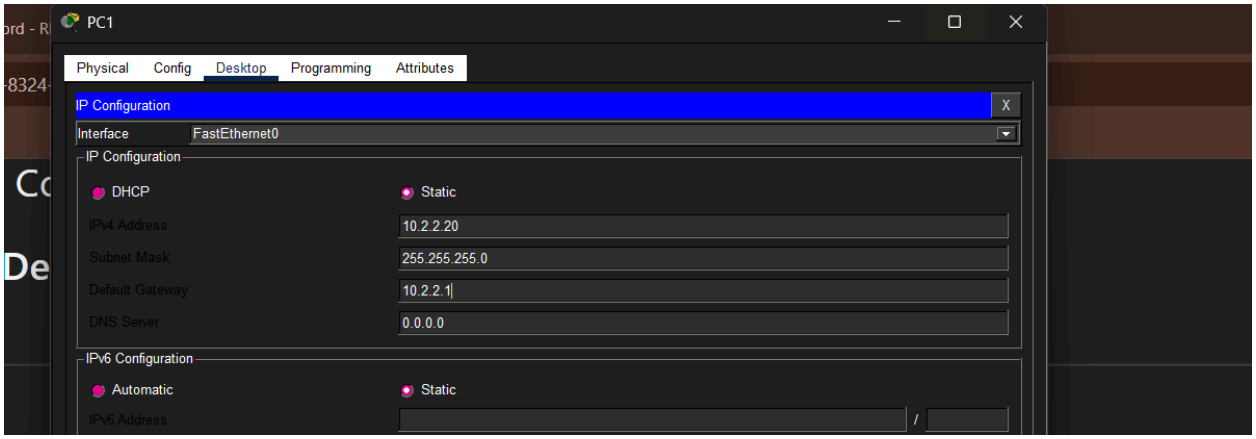
```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 20.1.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface gig0/1
Router(config-if)#ip address 10.2.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network id 20.1.1.0
                             ^
% Invalid input detected at '^' marker.

Router(config-router)#network 20.1.1.0
Router(config-router)#network 10.2.2.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

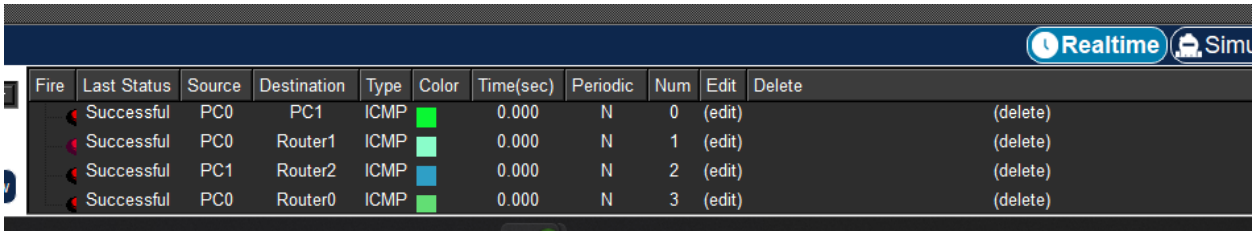
Pc 0-


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.2.10
Subnet Mask
255.255.0.0
Default Gateway
172.16.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static

Pc1-


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.2.2.20
Subnet Mask
255.255.255.0
Default Gateway
10.2.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address

o/p



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	Router1	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC1	Router2	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC0	Router0	ICMP		0.000	N	3	(edit)	(delete)

15-04-26 - OSPF

WILD CARD MASK

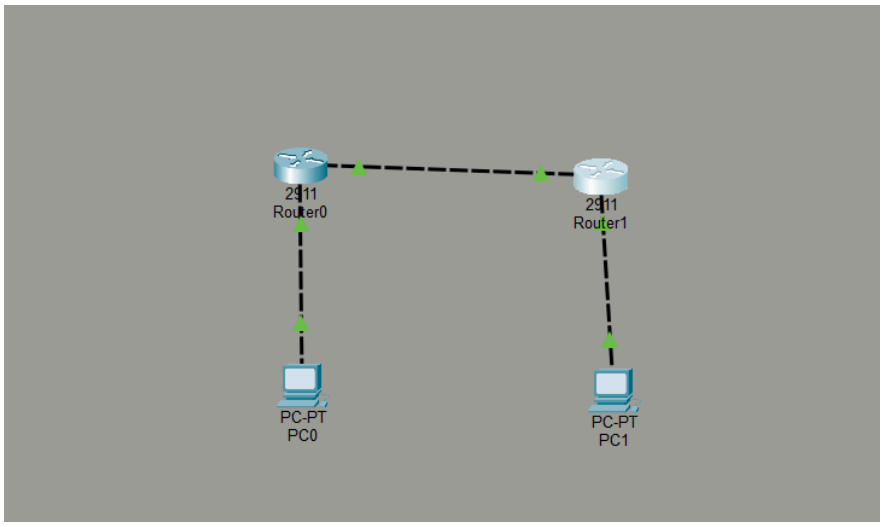
Network 192.168.5.0

If subnet = 255.255.255.0 , make 0 = 1 and 1 = 0

I.e = 0.0.0.255

10.1.1.0/30 = 255.255.252.0

For 10.1.1.0 , subnet = 255.255.252.0 i.e = (Wild card mask) 0.0.0.3



Router 0-

Router0

Physical | Config | CLI | Attributes

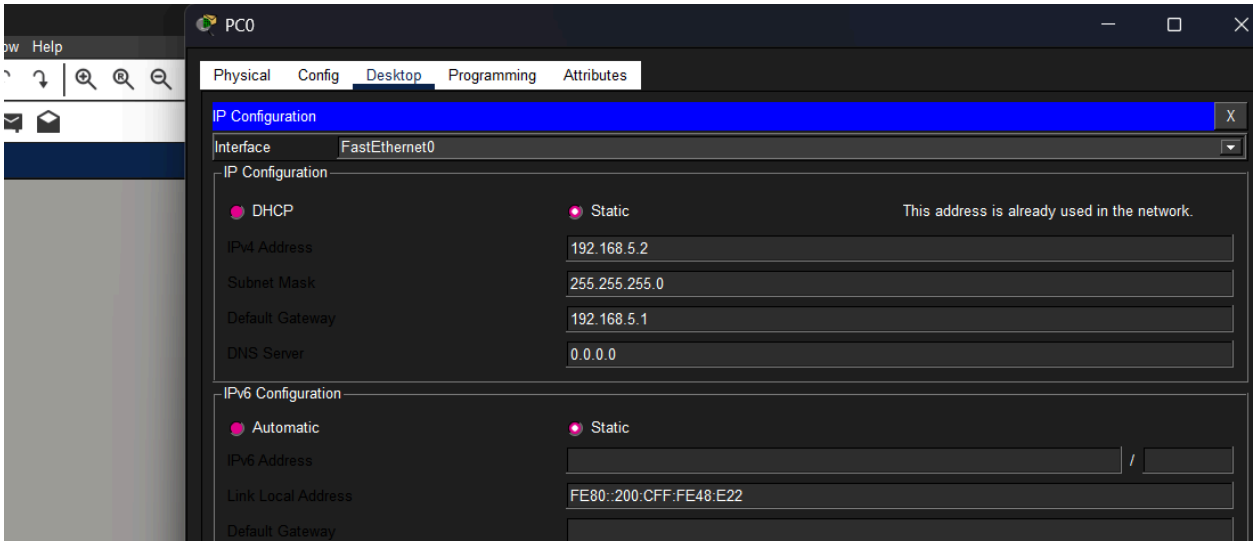
```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip interface gig0/0
                  ^
% Invalid input detected at '^' marker.

Router(config)#interface 0/0
                         ^
% Invalid input detected at '^' marker.

Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/0
Router(config-if)#ip address 10.1.1.1 255.255.252.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/1 192.168.5.1
                                ^
% Invalid input detected at '^' marker.

Router(config)#interface gig0/1
Router(config-if)#ip address 192.168.5.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%IP-4-DUPADDR: Duplicate address 192.168.5.1 on GigabitEthernet0/1, sourced by 0000.0C48.0E22

Router(config)#config t
%Invalid hex value
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 192.168.5.0
% Incomplete command.
Router(config-router)#network 192.168.5.0 0.0.0.255 area 0
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

00:10:44: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
```

Copy | Paste

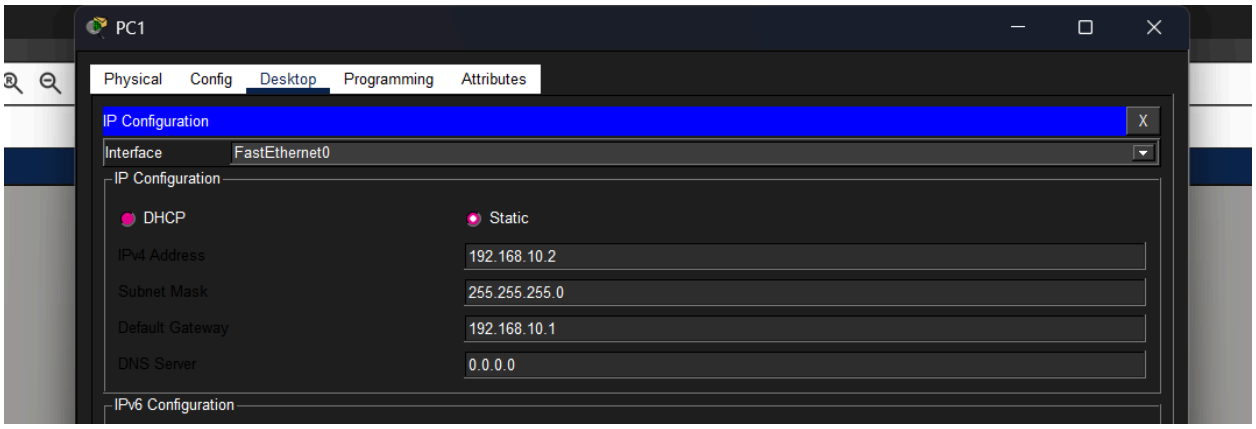
Top

Search

pc0-



Pc1



Router 1

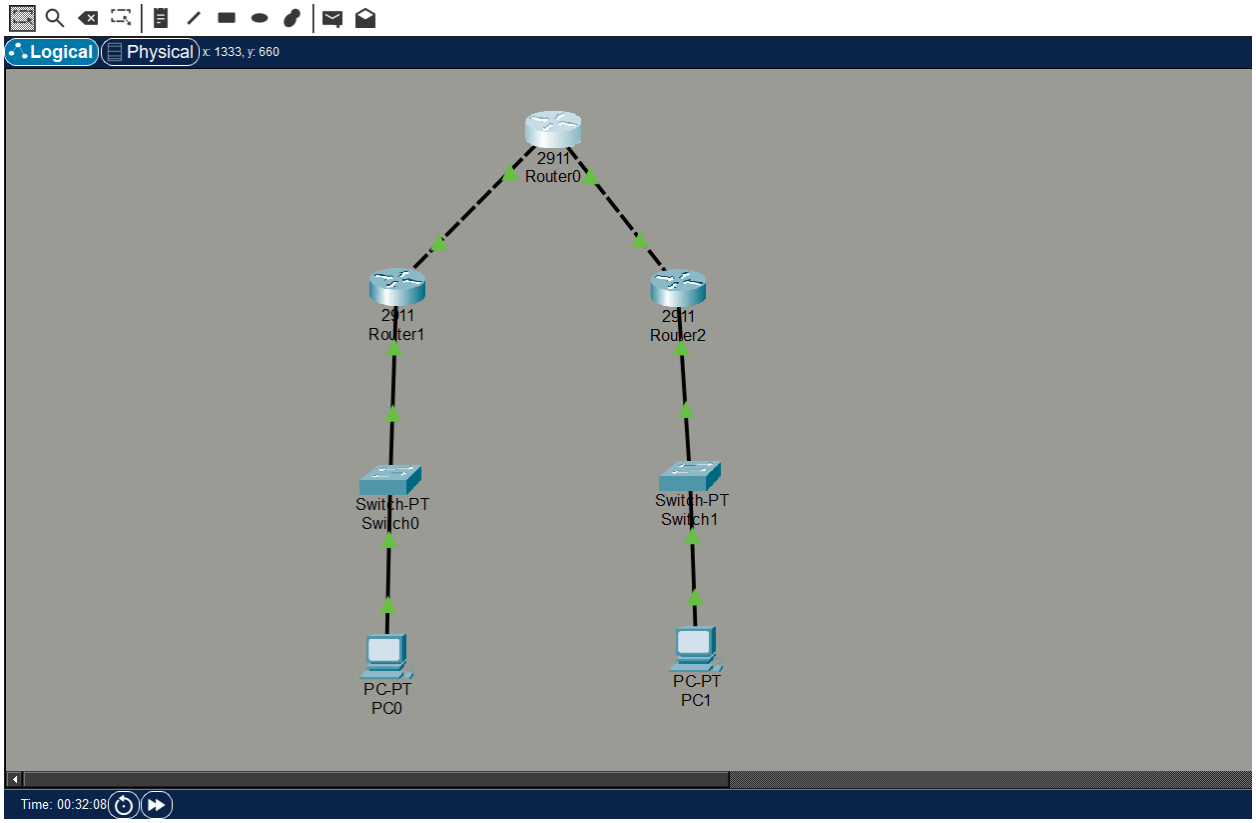
Router1

Physical Config CLI Attributes

```
         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 10.1.1.2 255.255.252.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gig0/1
Router(config-if)#ip address 192.168.10.1
% Incomplete command.
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
Router(config)#config t
%Invalid hex value
Router(config)#router ospf 1
Router(config-router)#router 2.2.2.2
Router(config-router)#router-id 2.2.2.2
Router(config-router)#network id 192.168.10.0 0.0.0.255 area 0
                                 ^
% Invalid input detected at '^' marker.

Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

00:10:39: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

Copy Paste

ASSIGN -7


Logical
Physical
x: 1333, y: 660
2911
Router0
2911
Router1
2911
Router2
Switch-PT
Switch0
Switch-PT
Switch1
PC-PT
PC0
PC-PT
PC1
Time: 00:32:08

Router 0

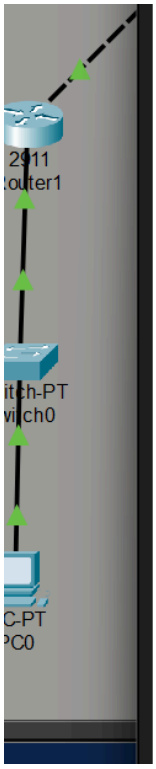
Ip ADD - 100.1.1.0 below not 10.1.1.0

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 10.1.1.0 255.255.255.0
Bad mask /24 for address 10.1.1.0
Router(config-if)#ip address 10.1.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/1
Router(config-if)#ip address 20.1.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config)#config t
%Invalid hex value
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#100.1.1.0 0.0.0.255 area 0
                     ^
% Invalid input detected at '^' marker.

Router(config-router)#network 100.1.1.0 0.0.0.255 area 0
Router(config-router)#network 20.1.1.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

00:16:11: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
```

Router 1

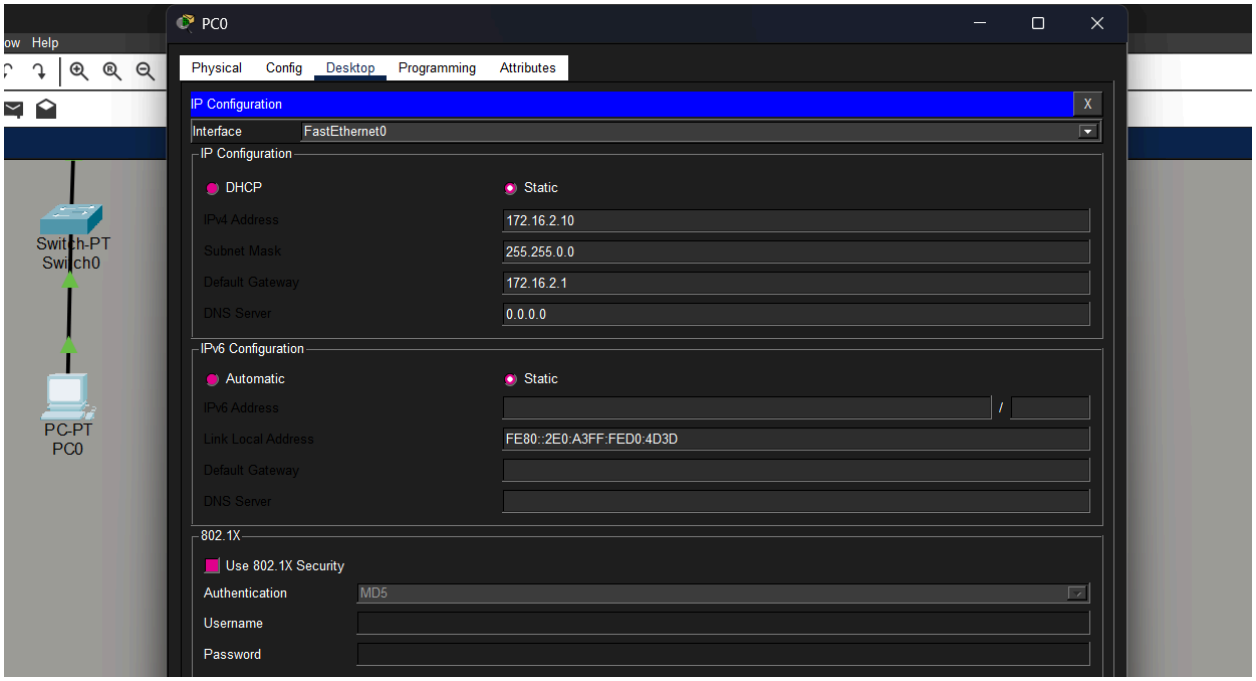
```
Router>
Router>
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 100.1.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#
% Unknown command or computer name, or unable to find computer address

Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/1
Router(config-if)#ip address 172.16.2.1 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#config t
%Invalid hex value
Router(config)#router-id 2.2.2.2
                        ^
% Invalid input detected at '^' marker.

Router(config)#router ospf 1
Router(config-router)#router-id 2.2.2.2
Router(config-router)#network 100.1.1.0 0.0.0.255 area 0
Router(config-router)#network 172.16.0.0 0.0.255.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 2



```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 20.1.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gig0/1
Router(config-if)#ip address 10.2.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#router ospf 1
Router(config-router)#router-id 3.3.3.3
Router(config-router)#network-id 20.1.1.0 0.0.0.255 area 0
                                         ^
% Invalid input detected at '^' marker.

Router(config-router)#network 20.1.1.0 0.0.0.255 area 0
Router(config-router)#network 10.2.2.0 0.0
00:16:05: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
.0
                                         ^
% Invalid input detected at '^' marker.

Router(config-router)#network 10.2.2.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

PC0-

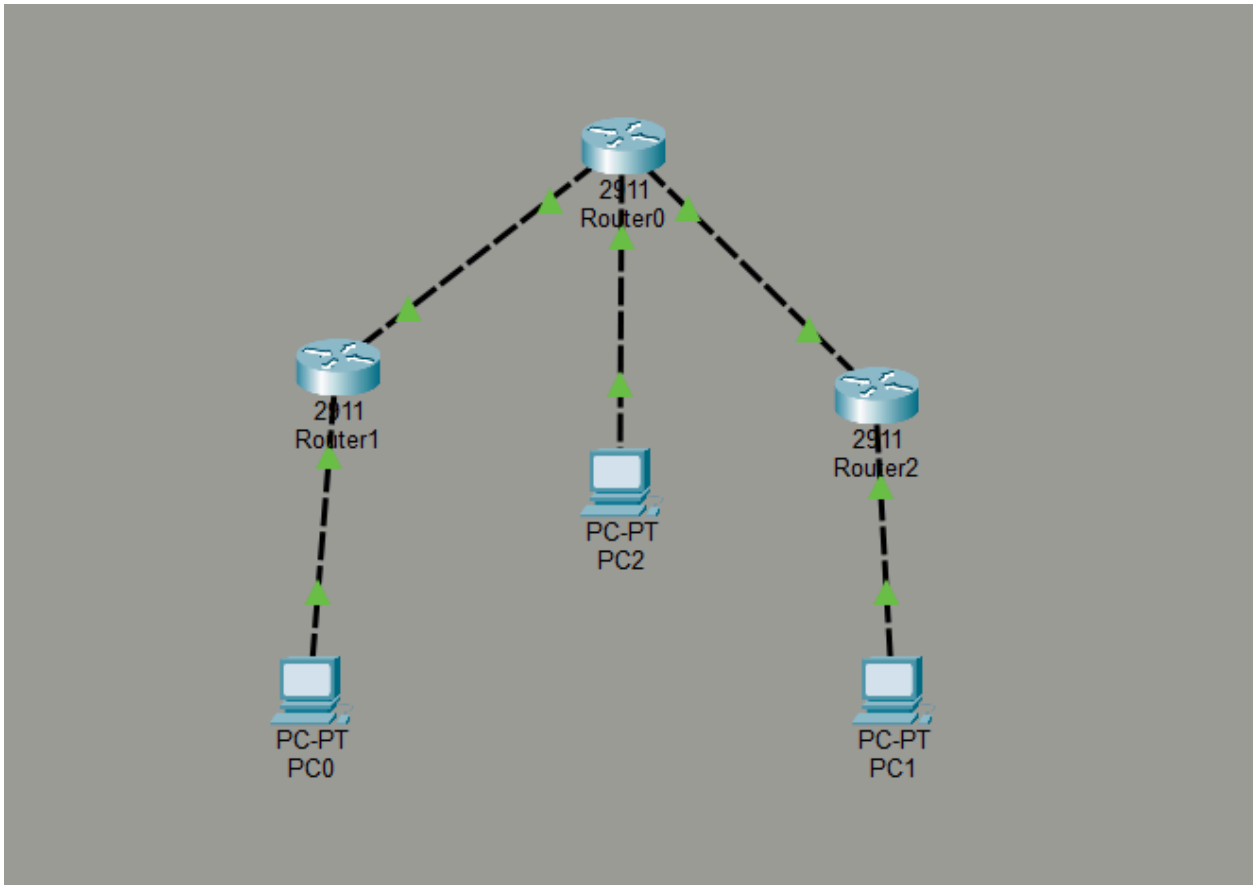


PC1-

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.2.2.20
Subnet Mask
255.0.0.0
Default Gateway
10.2.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static

ASSIGNMENT 8

Example -



Router -0

```
Router>
Router>enable
Router#interface gig0/0
                    ^
% Invalid input detected at '^' marker.

Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 172.16.0.2
% Incomplete command.
Router(config-if)#ip address 172.16.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/1
Router(config-if)#ip address 172.14.0.1 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#config t
%Invalid hex value
Router(config)#ip address 50.0.0.1 255.0.0.0
                    ^
% Invalid input detected at '^' marker.

Router(config)#interface gig0/2
Router(config-if)#ip address 50.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
exit
Router(config)#
```

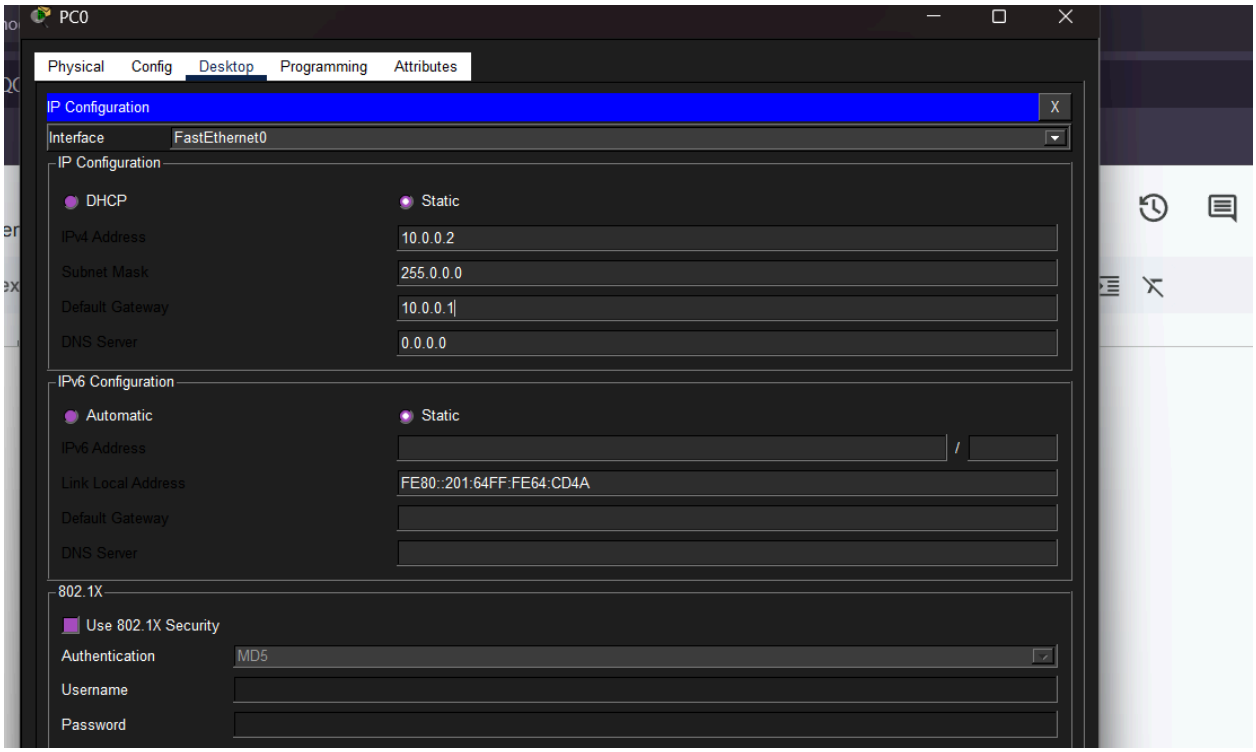
Router 1

```
Router>enable
Router#conig t
                ^
% Invalid input detected at '^' marker.

Router#
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 172.16.0.1 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit'
                      ^
% Invalid input detected at '^' marker.

Router(config-if)#exit
Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/1
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
```

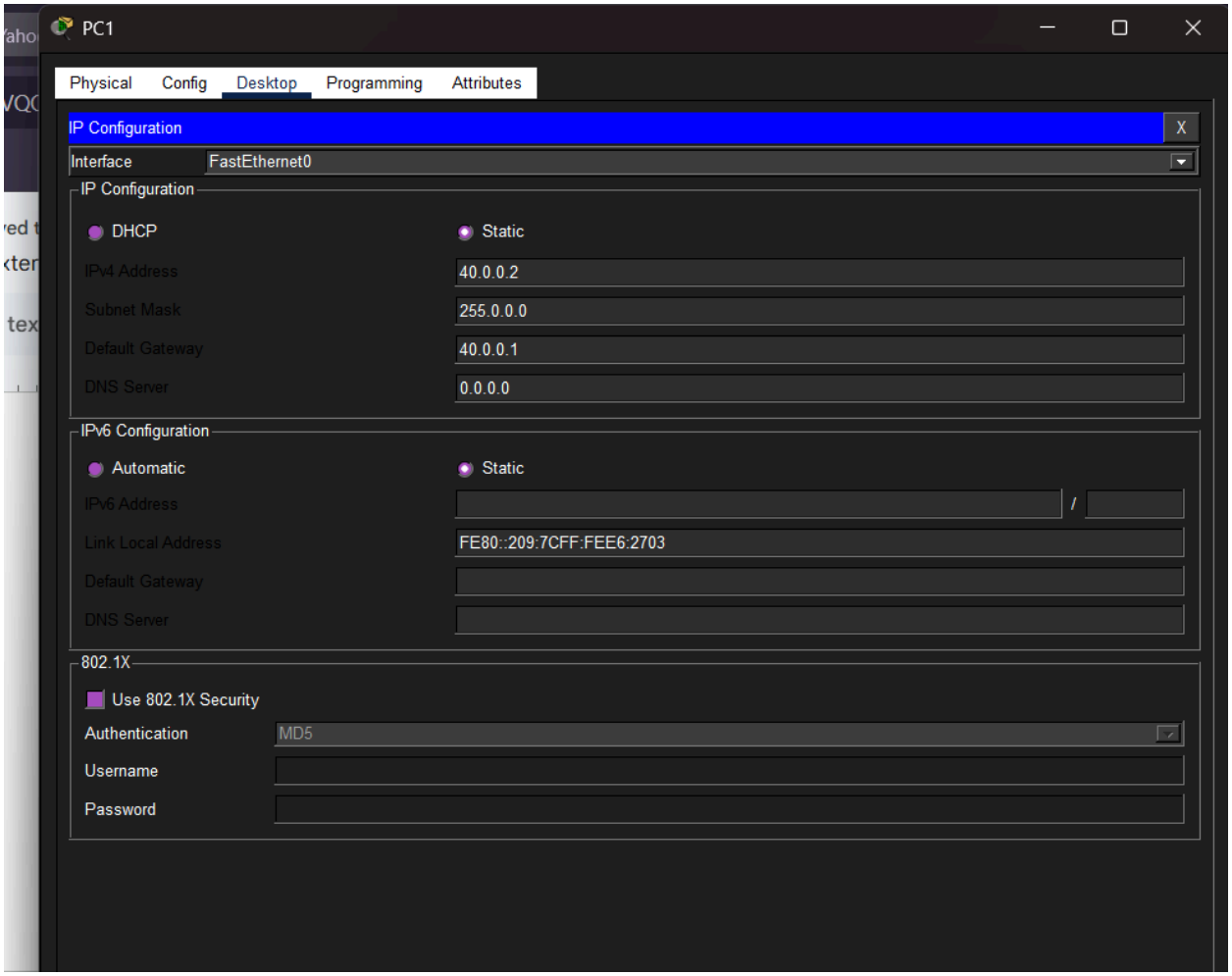
Router 2

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 172.14.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#
Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/1
Router(config-if)#ip address 40.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
```

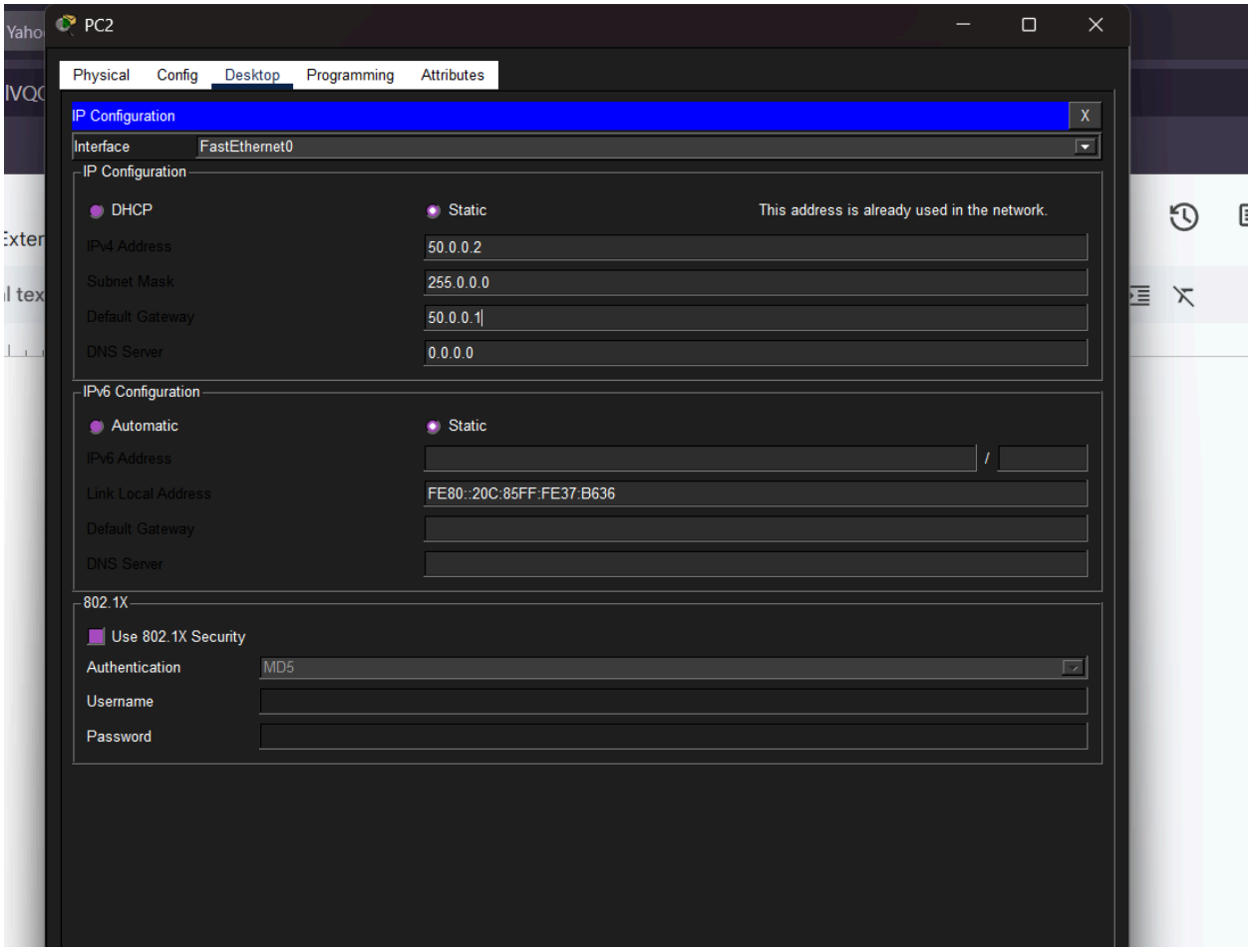
Pc0



Pc1


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
40.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
40.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::209:7CFF:FEE6:2703
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

Pc2-



BGP

Router 0 -cont..

```
Router(config)#router bgp 71
BGP is already running; AS is 1
Router(config)#no router bgp 1
Router(config)#router bgp 71
Router(config-router)#neighbor 172.16.0.1 remote as 1
                                                  ^
% Invalid input detected at '^' marker.

Router(config-router)#neighbor 172.16.0.1 remote-as 1
Router(config-router)#neighbor 172.14.0.2 remote-as 79
Router(config-router)#network 50.0.0.0 mask 255.0.0.0
Router(config-router)#exit
Router(config)#%BGP-5-ADJCHANGE: neighbor 172.16.0.1 Up
%BGP-5-ADJCHANGE: neighbor 172.14.0.2 Up
```

Router 1-cont..

```
Router>
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 1
Router(config-router)#neighbor 172.16.0.2 remote-as 71
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 172.16.0.2 Up
network
% Incomplete command.
Router(config-router)#network 10.0.0.0 mask 255.0.0.0
Router(config-router)#exit
Router(config)#
```

Copy

Router 2-cont..

```
Router>
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 79
Router(config-router)#neighbor 172.14.0.1 remote-as 71
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 172.14.0.1 Up

Router(config-router)#network 40.0.0.0 mask 255.0.0.0
Router(config-router)#exit
Router(config)#
```

Top

Output-

Realtime | Simulation

Scenario 0

New | Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC2	Router0	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC0	PC2	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC0	PC1	ICMP		0.000	N	4	(edit)	(delete)